

User Manual

for S32K1_S32M24X RM Driver

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1 Revision History	2
2 Introduction	3
2.1 Supported Derivatives	3
2.2 Overview	4
2.3 About This Manual	4
2.4 Acronyms and Definitions	5
2.5 Reference List	5
3 Driver	6
3.1 Requirements	6
3.2 Driver Design Summary	6
3.2.1 DMAMUX	6
3.3 Hardware Resources	7
3.4 Deviations from Requirements	7
3.5 Driver Limitations	8
3.6 Driver usage and configuration tips	8
3.6.1 HLD usage	8
3.6.1.1 API controls	8
3.6.2 LLD usage	8
3.6.3 Interface configuration	8
3.7 Runtime errors	10
3.8 Symbolic Names Disclaimer	10
3.9 Usage in non AUTOSAR context	10
4 Configuration Parameters	12
4.1 Rm	12
4.1.1 Module Rm	13
4.1.2 Container RmGeneral	14
4.1.3 Parameter Rm_VersionInfoApi	14
4.1.4 Parameter RmDevErrorDetect	14
4.1.5 Parameter RmEnableUserModeSupport	15
4.1.6 Parameter RmEnableDmaMuxSupport	15
4.1.7 Parameter RmEnableGlobalInit	16
4.1.8 Parameter RmMultiPartitionSupport	16
4.1.9 Reference RmInitEcucPartitionRef	17
4.1.10 Container RmConfigSet	17
4.1.11 Reference RmEcucPartitionRef	17
4.1.12 Container Dma_Mux_Configuration	18
4.1.13 Parameter Dma_Mux_DevErrorDetect	18
4.1.14 Container Dma_Mux_Module_Config	18
4.1.15 Parameter Dma_Mux_HwInstance	19
4.1.16 Parameter Dma_Mux_HwChannel	19
4.1.17 Parameter Dma_Mux_Enable_Trigger	20
4.1.18 Parameter Dma_Mux_Source0	20
4.1.19 Container CommonPublishedInformation	21
4.1.20 Parameter ArReleaseMajorVersion	21
4.1.21 Parameter ArReleaseMinorVersion	22
4.1.22 Parameter ArReleaseRevisionVersion	22
4.1.23 Parameter ModuleId	23
4.1.24 Parameter SwMajorVersion	23
4.1.25 Parameter SwMinorVersion	24
4.1.26 Parameter SwPatchVersion	24
4.1.27 Parameter VendorApiInfix	25

4.1.28 Parameter VendorId	25
5 Module Index	26
5.1 Software Specification	26
6 Module Documentation	27
6.1 RM Driver	27
6.1.1 Detailed Description	27
6.1.2 Macro Definition Documentation	27
6.1.2.1 RM_STOP_SEC_CONFIG_DATA_UNSPECIFIED	27
6.2 DMA_MUX IPV Driver	28
6.2.1 Detailed Description	28
6.2.2 Data Structure Documentation	28
6.2.2.1 struct Dma_Mux_Ip_ChannelConfigType	28
6.2.2.2 struct Dma_Mux_Ip_ConfigType	28

Chapter 1

Revision History

Revision	Date	Author	Description
1.0	07.03.2025	NXP RTD Team	S32K1_S32M24X Real-Time Drivers AUTOSAR R21-11 Version 3.0.0

Chapter 2

Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
- [Reference List](#)

This User Manual describes Resource Manager for S32K1_S32M24X. Resource Manager driver configuration parameters and deviations from the specification are described in Driver chapter of this document.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32k116_qfn32
- s32k116_lqfp48
- s32k118_lqfp48
- s32k118_lqfp64
- s32k142_lqfp48
- s32k142_lqfp64
- s32k142_lqfp100
- s32k142w_lqfp48
- s32k142w_lqfp64
- s32k144_lqfp48
- s32k144_lqfp64 / MWCT1014S_lqfp64
- s32k144_lqfp100 / MWCT1014S_lqfp100
- s32k144_mapbga100
- s32k144w_lqfp48
- s32k144w_lqfp64
- s32k146_lqfp64

- s32k146_lqfp100 / MWCT1015S_lqfp100
- s32k146_mapbga100 / MWCT1015S_mapbga100
- s32k146_lqfp144
- s32k148_lqfp100
- s32k148_mapbga100 / MWCT1016S_mapbga100
- s32k148_lqfp144
- s32k148_lqfp176
- s32m241_lqfp64
- s32m242_lqfp64
- s32m243_lqfp64
- s32m244_lqfp64

All of the above microcontroller devices are collectively named as S32K1_S32M24X. Note: MWCT part numbers contain NXP confidential IP for Qi Wireless Power

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
ASM	Assembler
BSMI	Basic Software Make file Interface
C/CPP	C and C++ Source Code
CDD	Complex Device Driver
DEM	Diagnostic Event Manager
DET	Development Error Tracer
DMA MUX	Direct Memory Access Multiplexer
ECU	Electronic Control Unit
LSB	Least Significant Bit
MCU	Micro Controller Unit
MIDE	Multi Integrated Development Environment
MRC	Memory Region Controller
MSB	Most Significant Bit
N/A	Not Applicable
RAM	Random Access Memory
RM	Resource Manager
SIU	Systems Integration Unit
SWS	Software Specification
XML	Extensible Markup Language

2.5 Reference List

#	Title	Version
1	S32K1xx Reference Manual	Rev.14.1, 01/2024
2	S32M24x Reference Manual	Rev. 5, 12/2024
3	S32K1xx Data Sheet	Rev. 14, 08/2021
4	S32M2xx Data Sheet	Rev. 7, 12/2024
5	S32K116 Errata	Mask Set Errata for Mask 0N96V Rev. 17/JAN/2024, 1/2024
6	S32K118 Errata	Mask Set Errata for Mask 0N97V Rev. 15/JAN/2024, 1/2024
7	S32K142 Errata	Mask Set Errata for Mask 0N33V Rev. 15/JAN/2024, 1/2024
8	S32K144 Errata	Mask Set Errata for Mask 0N57U Rev. 15/JAN/2024, 1/2024
9	S32K144W Errata	Mask Set Errata for Mask 0P64A Rev. 03/JUL/2024, 7/2024
10	S32K146 Errata	Mask Set Errata for Mask 0N73V Rev. 15/JAN/2024, 1/2024
11	S32K148 Errata	Mask Set Errata for Mask 0N20V Rev. 15/JAN/2024, 1/2024
12	S32M242 Errata	Mask Set Errata for Mask N33V+P69K Rev.1, 8/2024
13	S32M244 Errata	Mask Set Errata for Mask P64A+P69K Rev.1, 8/2024

Chapter 3

Driver

- [Requirements](#)
- [Driver Design Summary](#)
- [Hardware Resources](#)
- [Deviations from Requirements](#)
- [Driver Limitations](#)
- [Driver usage and configuration tips](#)
- [Runtime errors](#)
- [Symbolic Names Disclaimer](#)
- [Usage in non AUTOSAR context](#)

3.1 Requirements

Resource Manager is a Complex Device Driver (CDD), so there are no AUTOSAR requirements regarding this module.

It has vendor-specific requirements and implementation.

3.2 Driver Design Summary

The RM module provides a way to initialize and control the resource domains allocation and memory protection on the chip with the supported peripherals.

3.2.1 DMAMUX

The Direct Memory Access Multiplexer (DMAMUX) routes DMA sources, called slots, to any of the DMA channels. The RM DMAMUX driver provides a quick way to configure DMAMUX with DMA source and DMA channel selections.

#	Chip	Number of DMAMUX channels
1	S32K116	4
2	S32K118	4
3	S32K142	16
4	S32K142W	16

#	Chip	Number of DMAMUX channels
5	S32K144	16
6	S32K144W	16
7	S32K146	16
8	S32K148	16
9	S32M24x	16

The below image show DMAMUX block diagram:

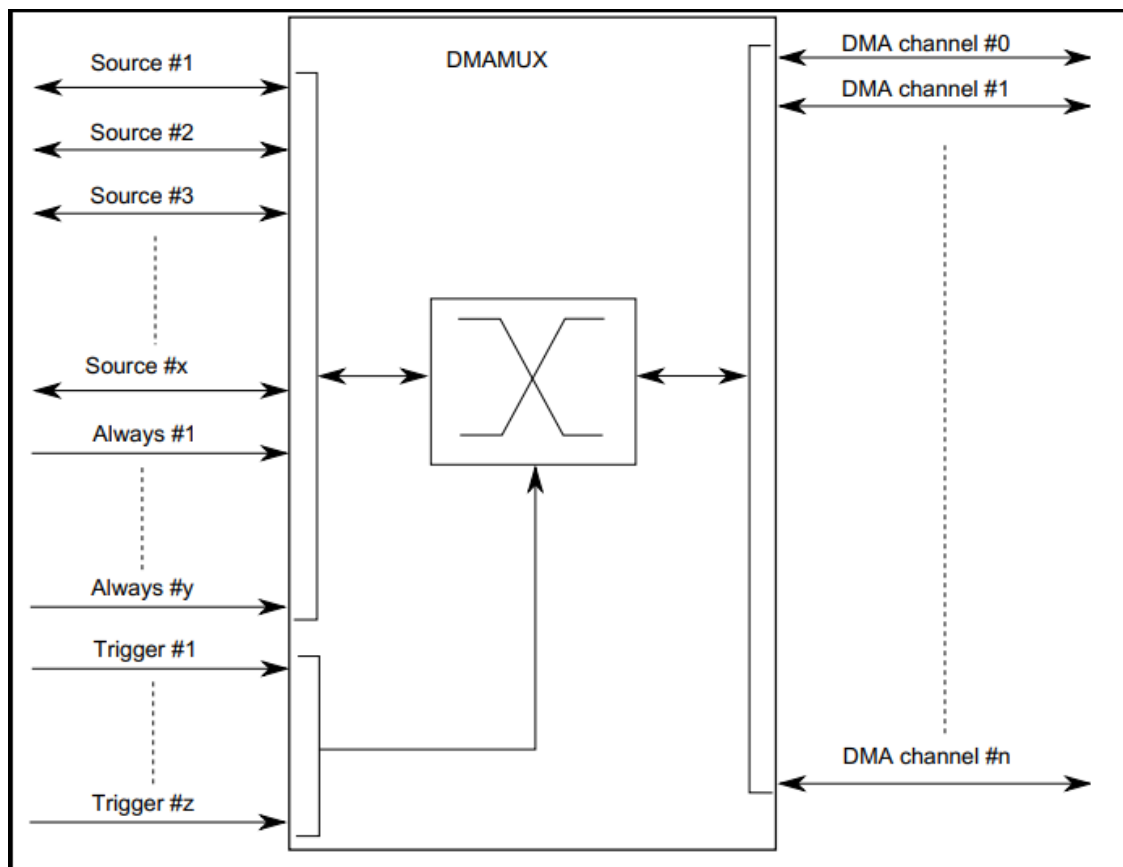


Figure 3.1 DMAMUX block diagram

3.3 Hardware Resources

#	Hardware IP	Description
1	DMA MUX	DMA Multiplexer

3.4 Deviations from Requirements

Since this is a CDD Module, there are no AUTOSAR requirements for the functionality.

Term	Definition
N/S	Out of scope

Term	Definition
N/I	Not implemented
N/F	Not fully implemented

Table 3.4 Driver Deviations Table

Requirement	Status	Description	Notes
N/A	N/A	N/A	N/A

3.5 Driver Limitations

- None

3.6 Driver usage and configuration tips

3.6.1 HLD usage

Prior usage of the RM CDD in an application, the configuration files must be generated with the configurator.

3.6.1.1 API controls

Initialization of all hardware resources is done using `Rm_Init`. Considering that the function is configuring domains and access to resources it is recommended that it is called before other bus masters are active, or that they are not accessing those resources.

3.6.2 LLD usage

RM CDD does not support LLD.

3.6.3 Interface configuration

The user can choose enable / disable checkbox to use the corresponding modules.

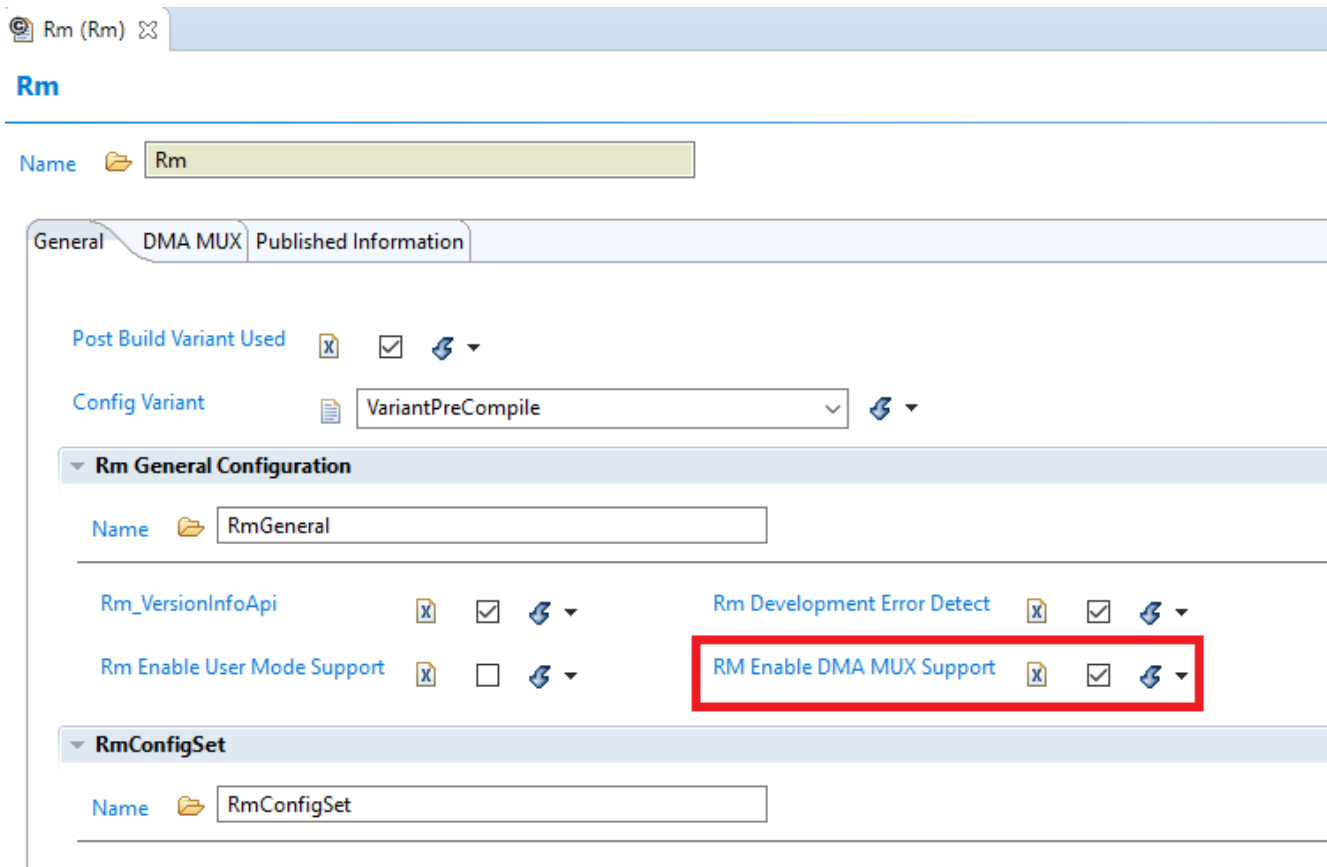


Figure 3.2 RM general config

DMAMUX configuration allows user routing the DMA source to DMA channel.

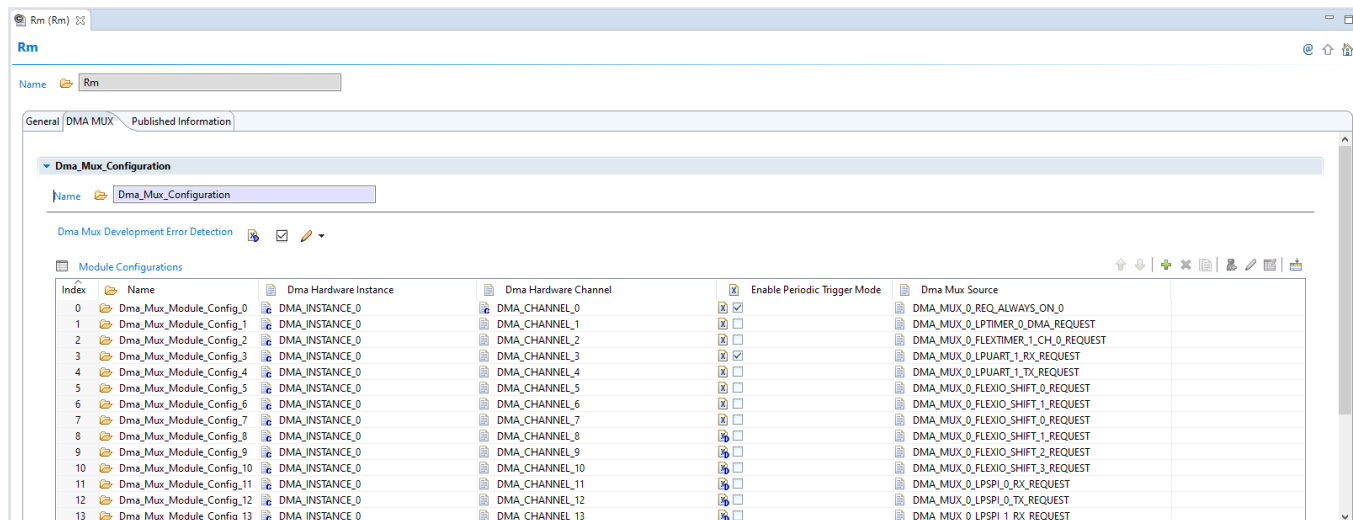


Figure 3.3 RM DMAMUX configuration

The triggers from TRGMUX module can trigger a DMA transfer on the first four DMA channels, for example, the LPIT can trigger DMA via TRGMUX. The user can enable this feature with "Enable Periodic Trigger Mode"

column.

3.7 Runtime errors

- Development Error Description

Error Code	Value	Condition triggering the error
RM_E_INIT_FAILED_U8	4	Rm_Init() is called with input parameter is invalid
RM_E_ALREADY_INITIALIZED_U8	2	Rm_Init() is called when the RM module is being initialized
RM_E_PARAM_POINTER	5	Rm_GetVersionInfo() is called with input parameter is NULL_PTR
RM_E_PARAM_CONFIG	7	Rm_Init() is called from a wrong partition.

3.8 Symbolic Names Disclaimer

All containers having symbolicNameValue set to TRUE in the AUTOSAR schema will generate defines like: `#define <Mip>Conf_<Container_ShortName>_<Container_ID>`

For this reason it is forbidden to duplicate the names of such containers across the RTD configurations or to use names that may trigger other compile issues (e.g. match existing `#ifdefs` arguments).

3.9 Usage in non AUTOSAR context

For the non-ASR environments, both High-Level and IP layer interfaces are exposed and can be used by software units above. These may consist of either higher level NXP SW deliverables, like stacks and middleware, or the application code itself.

The goal of the product is to be versatile, easy to use and integrate with other SW environments, offering both standardized and optimized low-level interfaces.

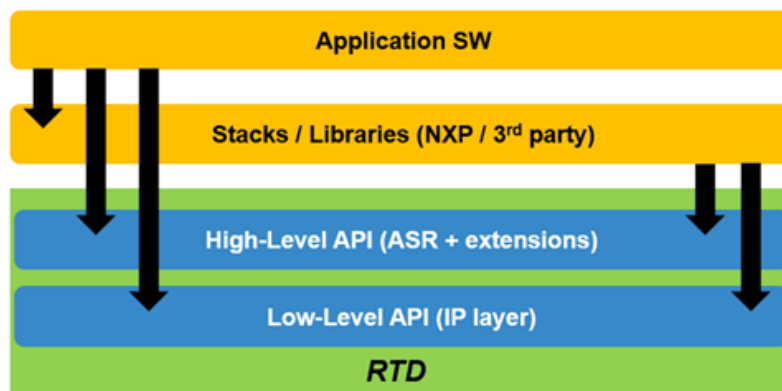


Figure 3.4 RTD Usage in non-ASR Context

In the non-AUTOSAR environment, there are no specific ECU states defined, but an external assumption for the application is that the correct sequence of initialization takes place prior to calling any runtime APIs.

The constraints considered for defining the APIs in the non-AUTOSAR environment are:

- Hardware resources used by the drivers must be properly initialized prior to runtime usage by calling the initialization function with configuration parameters generated by the configuration tools
- All dependencies must be initialized prior to the usage of a driver that depends on them; this translates to a general initialization sequence applicable for all applications, as follows:
 1. Initialization of the device clock tree so other periphery can be initialized
 2. Configuration of the pinout so peripherals with external connections can communicate with the outside world
 3. Configuration of the common resources (e.g. DMA channels, memory regions, access rights, etc) prior to the usage of any runtime functionality that has to benefit from this overall SoC configuration
 4. Initialization of specific drivers
 5. Runtime functionality
 6. Deinitialization (inverse initialization order)

High-level and IP layer functionalities cannot be used simultaneously over the same hardware resource. The granularity of the resource must be defined specifically for each driver, but as a general rule the default granularity considered is the peripheral instance (e.g. Gpt high level APIs cannot be called interchangeably with Pit_Ip low level APIs over the same PIT hardware instance).

The diagram below shows a generic, high-level sequence of calls to RTD interfaces in the non-AUTOSAR scenario:

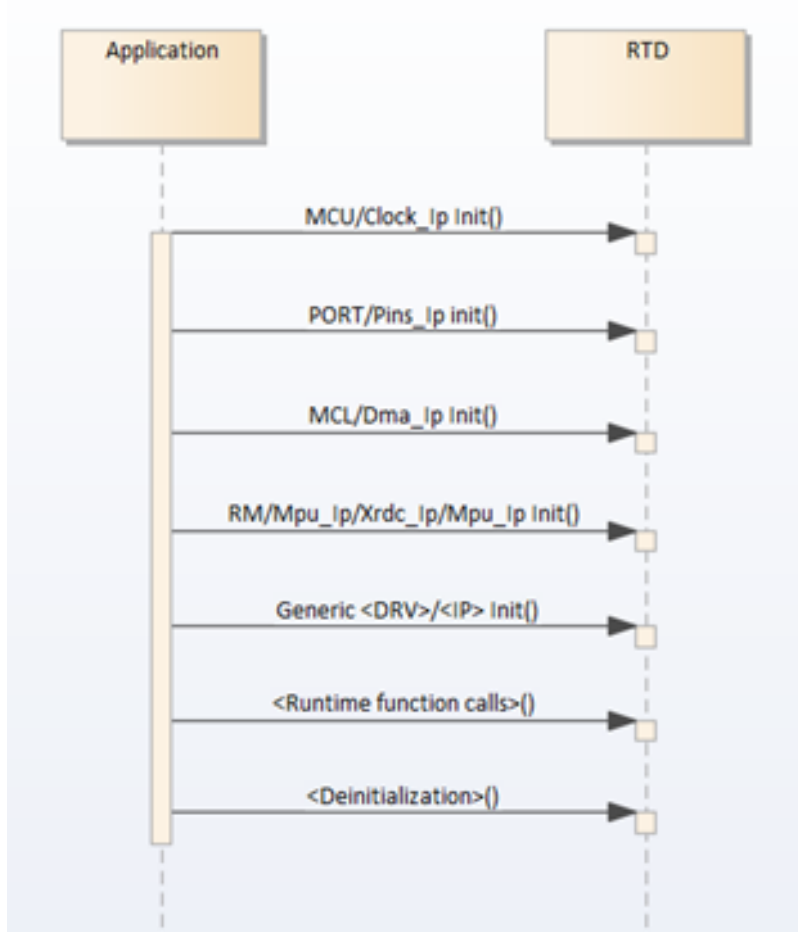


Figure 3.5 Non-AUTOSAR Sequence diagram

Chapter 4

Configuration Parameters

RTD provides configuration schema for the high level driver in EB Tresos proprietary format (**.xdm**), S32CT proprietary format (**.component**), but also in a standardized AUTOSAR format (**.epd**). For the low level driver configuration only the S32CT proprietary format (**.component**) is supported. The configuration data of the project can be saved in EB Tresos proprietary format (**.xdm**) and AUTOSAR standardized format (**.epc**) from EB Tresos and in S32CT proprietary format (**.mex**) and AUTOSAR standardized format (**.epc**) from S32 Design Studio. RTD provides templates for parsing the configuration data and generating configuration structures from it in XPath format, to be used with the EB Tresos generator and in JS format, to be used with the S32CT generator. Since the generation process is not standardized by AUTOSAR, only the two mentioned tools can be used with the provided templates. Using the same input configuration data, the same configuration structures will be obtained using any of the generators.

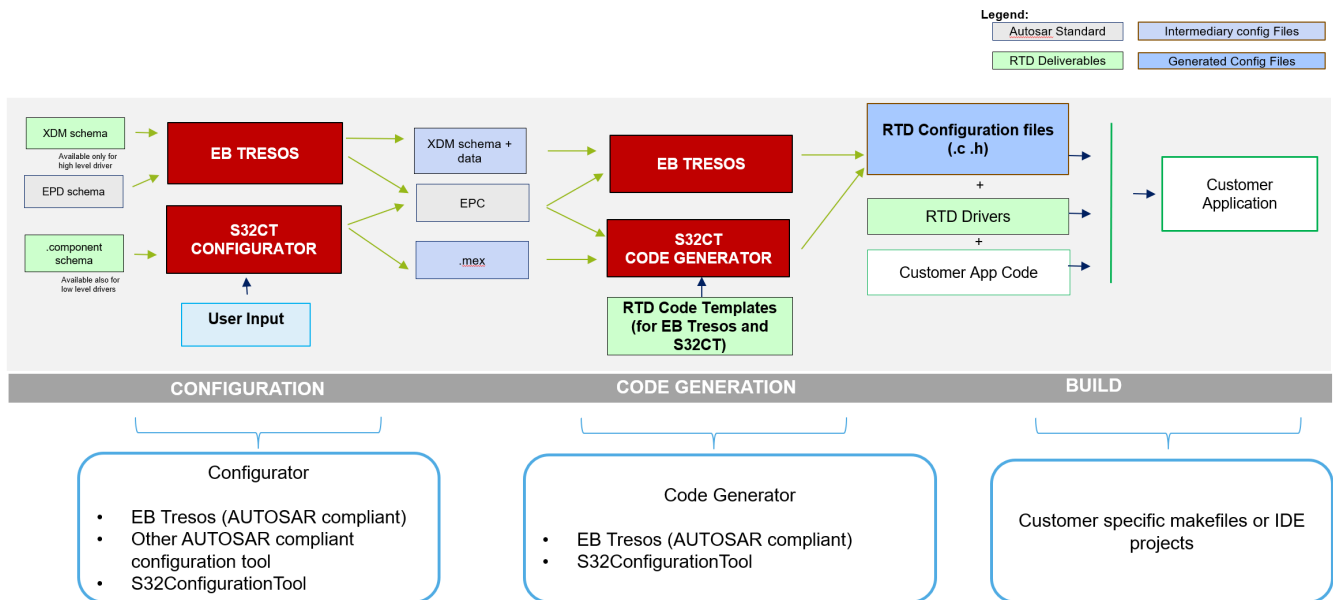


Figure 4.1 Configuration generation flow

- [Rm](#)

4.1 Rm

- Module [Rm](#)

- Container [RmGeneral](#)
 - * Parameter [Rm_VersionInfoApi](#)
 - * Parameter [RmDevErrorDetect](#)
 - * Parameter [RmEnableUserModeSupport](#)
 - * Parameter [RmEnableDmaMuxSupport](#)
 - * Parameter [RmEnableGlobalInit](#)
 - * Parameter [RmMultiPartitionSupport](#)
 - * Reference [RmInitEcucPartitionRef](#)
- Container [RmConfigSet](#)
 - * Reference [RmEcucPartitionRef](#)
 - * Container [Dma_Mux_Configuration](#)
 - Parameter [Dma_Mux_DevErrorDetect](#)
 - Container [Dma_Mux_Module_Config](#)
 - Parameter [Dma_Mux_HwInstance](#)
 - Parameter [Dma_Mux_HwChannel](#)
 - Parameter [Dma_Mux_Enable_Trigger](#)
 - Parameter [Dma_Mux_Source0](#)
- Container [CommonPublishedInformation](#)
 - * Parameter [ArReleaseMajorVersion](#)
 - * Parameter [ArReleaseMinorVersion](#)
 - * Parameter [ArReleaseRevisionVersion](#)
 - * Parameter [ModuleId](#)
 - * Parameter [SwMajorVersion](#)
 - * Parameter [SwMinorVersion](#)
 - * Parameter [SwPatchVersion](#)
 - * Parameter [VendorApiInfix](#)
 - * Parameter [VendorId](#)

4.1.1 Module Rm

Vendor specific: Configuration of the Rm (Resource Manager) module.

Included containers:

- [RmGeneral](#)
- [RmConfigSet](#)
- [CommonPublishedInformation](#)

Property	Value
type	ECUC-MODULE-DEF
lowerMultiplicity	1

Configuration Parameters

Property	Value
upperMultiplicity	1
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-POST-BUILD, VARIANT-PRE-COMPILE

4.1.2 Container RmGeneral

Vendor specific: Configuration of general Rm parameters.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.3 Parameter Rm_VersionInfoApi

Vendor specific: Enables/Disables the get version info API function

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.1.4 Parameter RmDevErrorDetect

Vendor specific:

Switches the Development Error Detection and Notification on or off.

true: Enabled.

false: Disabled.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.1.5 Parameter RmEnableUserModeSupport

When this parameter is enabled, the RM module will adapt to run from User Mode, with the following measures:

b) using 'call trusted function' stubs for all internal function calls that access registers requiring supervisor mode.

for more information, please see chapter 5.7 User Mode Support in IM

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.1.6 Parameter RmEnableDmaMuxSupport

When this parameter is enabled, the Dma Mux APIs are enabled to the user

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.1.7 Parameter RmEnableGlobalInit

Enable Global Init Support.

If the node is checked, Rm driver will support the API Rm_Init to initialize all IPs.

Otherwise Rm driver will support init functions separately for IPs.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.1.8 Parameter RmMultiPartitionSupport

This parameter globally enables the possibility to support multi-partition.

If this parameter is enabled, at least one EcucPartition needs to be defined (in all variants).

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.1.9 Reference RmInitEcucPartitionRef

Maps a config part of Rm_Init to ECUC partitions.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.1.10 Container RmConfigSet

Vendor specific: This container is the base for a multiple configuration set

Included subcontainers:

- [Dma_Mux_Configuration](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.11 Reference RmEcucPartitionRef

Maps the Rm driver to zero or multiple ECUC partitions to make the modules API available in this partition.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

Configuration Parameters

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/EcuC/EcuPartitionCollection/EcuPartition

4.1.12 Container Dma_Mux_Configuration

The Direct Memory Access Multiplexer (DMAMUX) routes DMA sources configuration.

Included subcontainers:

- [Dma_Mux_Module_Config](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.13 Parameter Dma_Mux_DevErrorDetect

Dma_Mux_DevErrorDetect

Switches the Development Error Detection and Notification ON or OFF.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.1.14 Container Dma_Mux_Module_Config

DMA MUX configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD

4.1.15 Parameter Dma_Mux_HwInstance

Vendor specific:

Select the Hardware DMA Instance.

NOTE: This is an Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	DMA_INSTANCE_0
literals	['DMA_INSTANCE_0']

4.1.16 Parameter Dma_Mux_HwChannel

Vendor specific:

Select the physical eDMA Channel.

NOTE: This is an Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Configuration Parameters

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	DMA_CHANNEL_0
literals	['DMA_CHANNEL_0', 'DMA_CHANNEL_1', 'DMA_CHANNEL_2', 'DMA_CHANNEL_3', 'DMA_CHANNEL_4', 'DMA_CHANNEL_5', 'DMA_CHANNEL_6', 'DMA_CHANNEL_7', 'DMA_CHANNEL_8', 'DMA_CHANNEL_9', 'DMA_CHANNEL_10', 'DMA_CHANNEL_11', 'DMA_CHANNEL_12', 'DMA_CHANNEL_13', 'DMA_CHANNEL_14', 'DMA_CHANNEL_15']

4.1.17 Parameter Dma_Mux_Enable_Trigger

Enables the periodic trigger capability for the triggered DMA channel.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.18 Parameter Dma_Mux_Source0

Dma Mux Source

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	DMA_MUX_0_REQ_DISABLED

Property	Value
literals	['DMA_MUX_0_REQ_DISABLED', 'DMA_MUX_0_LPUART_0_RX_REQUEST', 'DMA_MUX_0_LPUART_0_TX_REQUEST', 'DMA_MUX_0_LPUART_1_RX_REQUEST', 'DMA_MUX_0_LPUART_1_TX_REQUEST', 'DMA_MUX_0_FLEXIO_SHIFT_0_REQUEST', 'DMA_MUX_0_FLEXIO_SHIFT_1_REQUEST', 'DMA_MUX_0_FLEXIO_SHIFT_2_REQUEST', 'DMA_MUX_0_FLEXIO_SHIFT_3_REQUEST', 'DMA_MUX_0_LPSPI_0_RX_REQUEST', 'DMA_MUX_0_LPSPI_0_TX_REQUEST', 'DMA_MUX_0_LPSPI_1_RX_REQUEST', 'DMA_MUX_0_LPSPI_1_TX_REQUEST', 'DMA_MUX_0_FLEXTIMER_1_CH_0_REQUEST', 'DMA_MUX_0_FLEXTIMER_1_CH_1_REQUEST', 'DMA_MUX_0_FLEXTIMER_1_CH_2_REQUEST', 'DMA_MUX_0_FLEXTIMER_1_CH_3_REQUEST', 'DMA_MUX_0_FLEXTIMER_1_CH_4_REQUEST', 'DMA_MUX_0_FLEXTIMER_1_CH_5_REQUEST', 'DMA_MUX_0_FLEXTIMER_1_CH_6_REQUEST', 'DMA_MUX_0_FLEXTIMER_1_CH_7_REQUEST', 'DMA_MUX_0_FLEXTIMER_0_CH_0_7_REQUEST', 'DMA_MUX_0_ADC_0_DMA_REQUEST', 'DMA_MUX_0_LPI2C_0_RX_REQUEST', 'DMA_MUX_0_LPI2C_0_TX_REQUEST', 'DMA_MUX_0_SS_L_PDB_PIONEER_0_REQUEST', 'DMA_MUX_0_ANALOG_COMPARATOR_0_REQUEST', 'DMA_MUX_0_RGPIOC_PCTL_PORTA_REQUEST', 'DMA_MUX_0_RGPIOC_PCTL_PORTB_REQUEST', 'DMA_MUX_0_RGPIOC_PCTL_PORTC_REQUEST', 'DMA_MUX_0_RGPIOC_PCTL_PORTD_REQUEST', 'DMA_MUX_0_RGPIOC_PCTL_PORTE_REQUEST', 'DMA_MUX_0_FLEXCAN_0_DMA_REQUEST', 'DMA_MUX_0_LPTIMER_0_DMA_REQUEST', 'DMA_MUX_0_REQ_ALWAYS_ON_0', 'DMA_MUX_0_REQ_ALWAYS_ON_1']

4.1.19 Container CommonPublishedInformation

Vendor specific:

Common container, aggregated by all modules. It contains published information about vendor and versions.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.20 Parameter ArReleaseMajorVersion

Vendor specific:

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Configuration Parameters

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.1.21 Parameter ArReleaseMinorVersion

Vendor specific:

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	7
max	7
min	7

4.1.22 Parameter ArReleaseRevisionVersion

Vendor specific:

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.1.23 Parameter ModuleId

Vendor specific:

Module ID of this module from Module List.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	255
max	255
min	255

4.1.24 Parameter SwMajorVersion

Vendor specific:

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION

Configuration Parameters

Property	Value
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	3
max	3
min	3

4.1.25 Parameter SwMinorVersion

Vendor specific:

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.1.26 Parameter SwPatchVersion

Vendor specific:

Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.1.27 Parameter VendorApiInfix

Vendor specific:

In driver modules which can be instantiated several times on a single ECU, BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name.

This parameter is used to specify the vendor specific name. In total, the implementation specific name is generated as follows: <ModuleName>_>VendorId>_<VendorApiInfix>.

E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a VendorApiInfix of "v11r456" a api name Can_Write defined in the SWS will translate to Can_123_v11r456Write.

This parameter is mandatory for all modules with upper multiplicity > 1. It shall not be used for modules with upper multiplicity =1.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	

4.1.28 Parameter VendorId

Vendor specific:

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	43
max	43
min	43



Chapter 5

Module Index

5.1 Software Specification

Here is a list of all modules:

RM Driver	27
DMA_MUX IPV Driver	28

Chapter 6

Module Documentation

6.1 RM Driver

6.1.1 Detailed Description

Macros

- `#define RM_STOP_SEC_CONFIG_DATA_UNSPECIFIED`

6.1.2 Macro Definition Documentation

6.1.2.1 RM_STOP_SEC_CONFIG_DATA_UNSPECIFIED

```
#define RM_STOP_SEC_CONFIG_DATA_UNSPECIFIED
```

Export RM configurations.

Definition at line 118 of file CDD_Rm.h.

6.2 DMA_MUX IPV Driver

6.2.1 Detailed Description

Data Structures

- struct [Dma_Mux_Ip_ChannelConfigType](#)
Configuration structure containing the channel configuration. [More...](#)
- struct [Dma_Mux_Ip_ConfigType](#)
IP configuration structure. [More...](#)

6.2.2 Data Structure Documentation

6.2.2.1 struct Dma_Mux_Ip_ChannelConfigType

Configuration structure containing the channel configuration.

Definition at line 82 of file Dma_Mux_Ip_Types.h.

Data Fields

uint8	Instance	Select the instance request source.
uint8	Channel	Select the channel request source.
uint8	ConfigValue	Configuration value for the channel.

6.2.2.2 struct Dma_Mux_Ip_ConfigType

IP configuration structure.

Definition at line 92 of file Dma_Mux_Ip_Types.h.

Data Fields

uint8	ChannelConfigCnt	Channel Count
const Dma_Mux_Ip_ChannelConfigType *	pChannelConfigArr	Channel configuration array

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